HDI Predictor Documentation

Brainstorming & Idea Prioritization

Date	10 June 2025
Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor – Human Development Index Classification System
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Step 1: Brainstorm and Idea Listing

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Sai Venkat	Build a classifier that predicts a country's HDI tier (Very High / High / Medium / Low) from its health, education, and income indicators	Core ML Model	1
2	Sai Venkat	Provide a regression estimate of the exact HDI score in addition to the tier classification	Enhancement	1
3	Sai Venkat	Let users compare two countries side-by-side to see which indicator is dragging their HDI down	User Insight	2
4	Sai Venkat	Deploy the model behind a simple Flask web form so policymakers/researchers can enter indicator values directly	Deployment	3
5	Sai Venkat	Containerize the whole app with Docker so it runs identically on any evaluator's machine	DevOps	3
6	Sai Venkat	Add an explainability layer (feature importance) so users understand *why* a country landed in a given tier	Enhancement	2

Step 2: Idea Prioritization

Group No.	Final Idea	Feasibility	Importance	Priority	Selected (Yes/No)
1	HDI tier classification using a RandomForestClassifier	High	High	1	Yes

	trained on life expectancy, schooling, and GNI per capita				
3	Flask web app with Docker containerization for reproducible deployment	High	High	2	Yes
2	Feature-importance explainability for model predictions	Medium	Medium	3	Yes (stretch goal)

Outcome: The team converged on building a supervised classification model (HDI Predictor) that ingests the four core UNDP HDI dimensions — life expectancy, mean years of schooling, expected years of schooling, and GNI per capita (PPP) — and outputs both the predicted HDI category and the underlying score, exposed through a Flask-based web interface and packaged with Docker for portability.